

Build a simple low power DC-DC Boost Converter

Introduction to the course:	In applications such as battery charger, battery power systems, power amplifier applications etc. Boost conversion is required from low voltage to high voltage. This project intends to provide a hands-on approach to build a low power dc-dc boost converter.
Contents	• To introduce circuit simulation using LT-spice software. • To create a dc-dc boost converter circuit using breadboard/dot board and test with a load(fan).
Methodology	• Step 1 - Circuit Simulation of the dc-dc boost converter using LT-spice. • Step 2 - Actual component testing on breadboard/dot board. With the required input supply, the output results are monitored using multimeter.
Course Prerequisites	• Understanding of boost converter • Basics of RLC circuits • Soldering (optional)

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Part 1 – Circuit Simulation using LT-Spice

Part 2 – Physical design of the circuit

Build a simple low power DC-DC Boost Converter using LT-Spice

Part – 1 Circuit Simulation using LT-Spice

Aim

Basic understanding about LT-Spice

LT-Spice simulation installation instructions

Connections - Circuit Diagram - Detailed Steps

Task - Exercise

Fundamental learning

Topic	Resource
Concept of Boost-Converter	YouTube link
Components & Working of Dc-Dc Boost converter	YouTube link

Aim

Very often a low power boost converter used to convert some low voltage to a higher voltage is required in a system. This build exercise is to learn to build one such simple converter, using LT-Spice simulation in an open loop.

LT-Spice - Introduction

LT-Spice is an open source, high-performance simulation tool developed by Analog Devices. It is used to simulate, design, and test electronic circuits before physically building them.

- **Why LT-Spice is important? (Theory)**

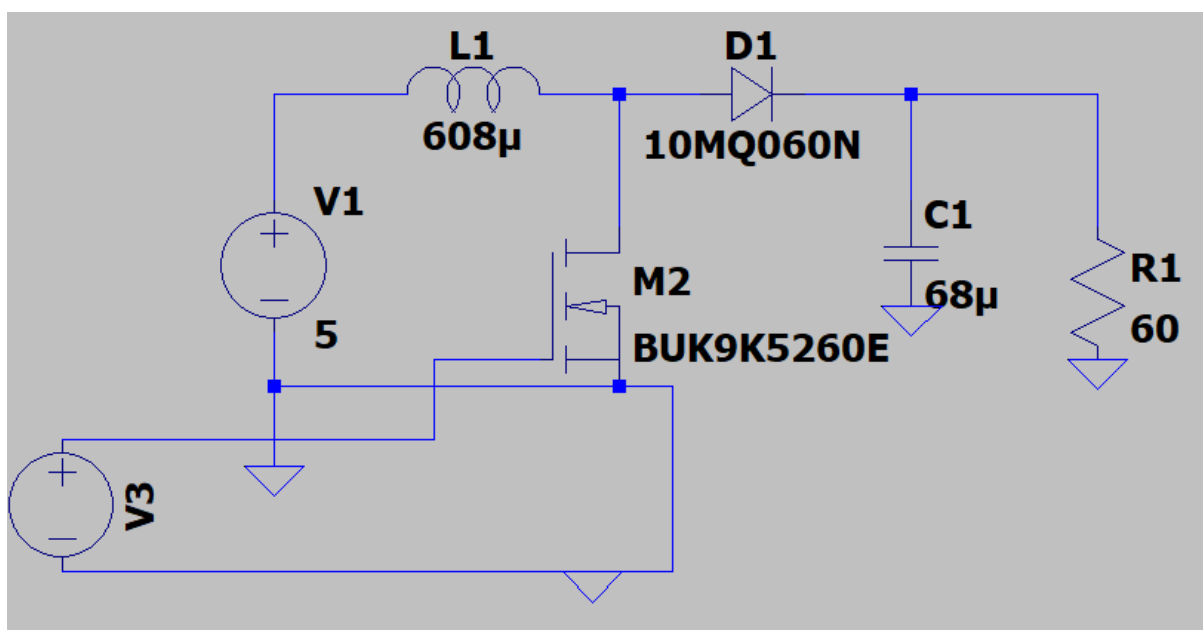
LT-Spice provides a safe learning environment where peoples can freely experiment with electronic circuits without risk of damaging physical components. It helps them to understand circuit behaviour by allowing real time visualization of voltage, current, and other parameters. By simulating before building, Peoples save time and cost as they as they do not need to buy multiple components or PCBs initially. LT-Spice also encourages peoples to develop strong problem -solving skills, teaching them how to identify and fix circuit issues logically. It offers valuable hands-on simulation experience that connects theoretical concepts learned in classrooms with practical real-world electronics. Most importantly, learning LT-Spice gives peoples and industry-relevant skill, preparing them for careers in research, development, and engineering industries.

LT-Spice Simulation Installation Instructions

- 1) **Open your web browser** (Such as Chrome, Firefox, or Edge).
- 2) In the search bar, type **LT-Spice** and press Enter.
- 3) Click on the official link from **Analog devices**, usually titled something like “**LTspice**” | **Analog devices**”. Or use this hyperlink to download [LT-Spice](#).
- 4) **On the LTspice page**, look for a **Download** button or section.
- 5) **Choose the correct version** for your operating system (Windows, macOS, etc.).
- 6) **Download the installer file** to your computer.
- 7) **Once the download is complete**, open the installer file.
- 8) **Follow the on-screen instructions** to install LT-Spice:
 - Accept the License Agreement
 - Choose the installation folder (or use the default)
 - Complete the installation process.
- 9) **After installation, open LT-Spice** from your desktop or Start Menu.
- 10) **Now are you ready to start using LT-Spice for circuit simulation.**

Connections

Circuit Diagram:



Note: After selecting a component, **click** on the schematic area to **place it**.

Step 4: Place Two Voltage Sources

- **V1: (Input 5V DC Supply)**
- **V3: (PWM gate signal for MOSFET)**

Note: In physical hardware setup we are using a component known as **Gate driver IC (TC4427)**, which gives the **PWM gate signal for MOSFET** to operate the duty cycle to achieve “**Boost Conversion**”. In LT-spice we can use **voltage source as a PWM gate signal for MOSFET**.

- **After placing the voltage sources, you can right-click on them to rename:**
- **Name first voltage source as V1.**
- **Name second voltage source as V3.**

Step 5: Arrange Components Properly

Arrange the components based on the circuit diagram for good understanding.

- 1) V1 (5V input) connect to inductor (L1).
- 2) Inductor (L1) connects to:
 - Anode (A) of diode (D1)
 - Drain (D) of MOSFET (M2)
- 3) Cathode (K) of diode (D1) connects to positive terminal of capacitor (C1)
- 4) Negative terminal of capacitor (C1) connects to ground (GND)
- 5) Capacitor (C1) and Resistor (R1) are connected in parallel (both connected across same points).
- 6) Source (S) of MOSFET (M2) connects to Ground (GND).
- 7) Gate (G) of MOSFET (M2) connects to PWM Voltage source (V3).

Step 6: Connect Wires

- Press F3 to press Start wiring
- Click on one pin of the component, drag, and click again to connect to another component.

Wire carefully

- Connect the components exactly as like in the circuit diagram.
- Connect the grounds properly wherever required.

Step 7: Component Values

Note: Right click each component to set the values:

Component	Value to set
V1 (voltage source)	DC = 5V
V3 (PWM source)	Will set as pulse source (explained below)
L1 (Inductor)	608 μH
D1(Diode)	Schottky diode (e.g., "10MQ060N")
M2 (MOSFET)	Choose "BUK9K5260E" or any standard NMOS
C1 (Capacitor)	68μF
R1 (Resistor)	60Ω

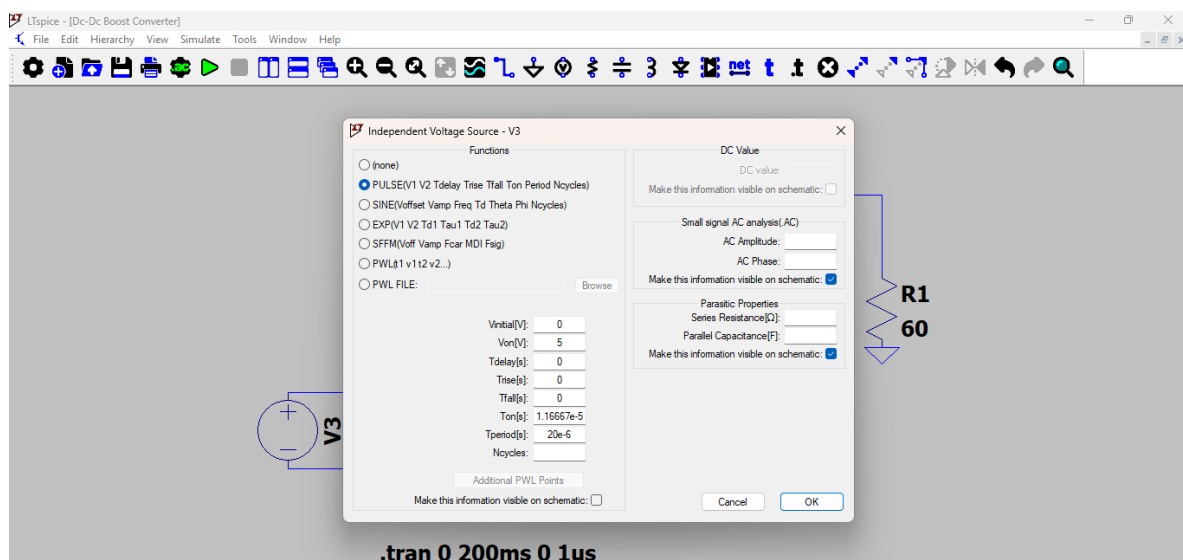
Step 8: Configure the PWM Signal (V3)

Note: This is very important for switching the MOSFET:

- Right click V3 (the second voltage)
- Select Advanced
- Choose PULSE and fill the below values.

Setting	Value
Vinitial	0V
Von	5V
Tdelay	0s
Ton	11.667 μ s
Tperiod	20 μ s
Ncycles	0(infinite)

Note: This will generate a PWM square wave of about 50KHz frequency.



Note: Use the above image for your references.

Understanding about Duty Cycle:

The Duty Cycle (%) tells us how long the MOSFET stays ON during one complete switching cycle.

Duty Cycle Formula:

$$\text{Duty Cycle(\%)} = \left(\frac{T_{on}}{T_{period}} \right) \times 100$$

$$\text{Duty Cycle} = (11/20) \times 100 = 55\%$$

Note: If you want to increase the Duty Cycle then increase the Ton period and observe the output.

Step 9: Add Ground Connections

- Every voltage source and component must have a ground connection!
- Press **P**, search for **GND**, and place it.
- Connect GND to:
 - 1) Negative side of V1 (input 5V).
 - 2) Negative side of V3 (PWM).
 - 3) Capacitor negative side.
 - 4) MOSFET source pin.

Step 10: Setup Simulation Command

- Press **S** or go to simulate > Edit simulation Cmd.
- Choose Transient tab.
- Set stop time:
 - 1) Stop time: 200ms
 - 2) Maximum Timestep: 1us
- Click Ok.
- Place the. **tran 0 200ms 1us** command on the schematic.

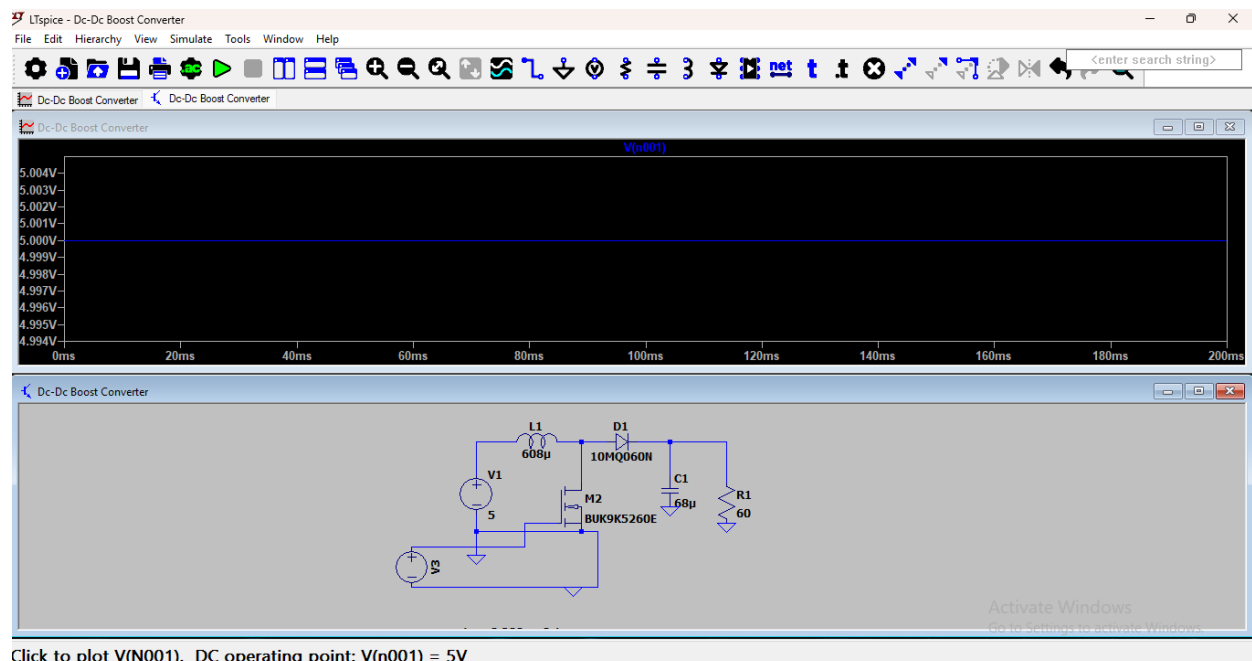
Step 11: Save and Run Simulation

- Click save.
- Then click the Run button.
- The simulation will start.
- You can probe:
 - 1) Voltage across the capacitor (C1).
 - 2) Current through Inductor (L1).
 - 3) Gate signal applied to MOSFET.

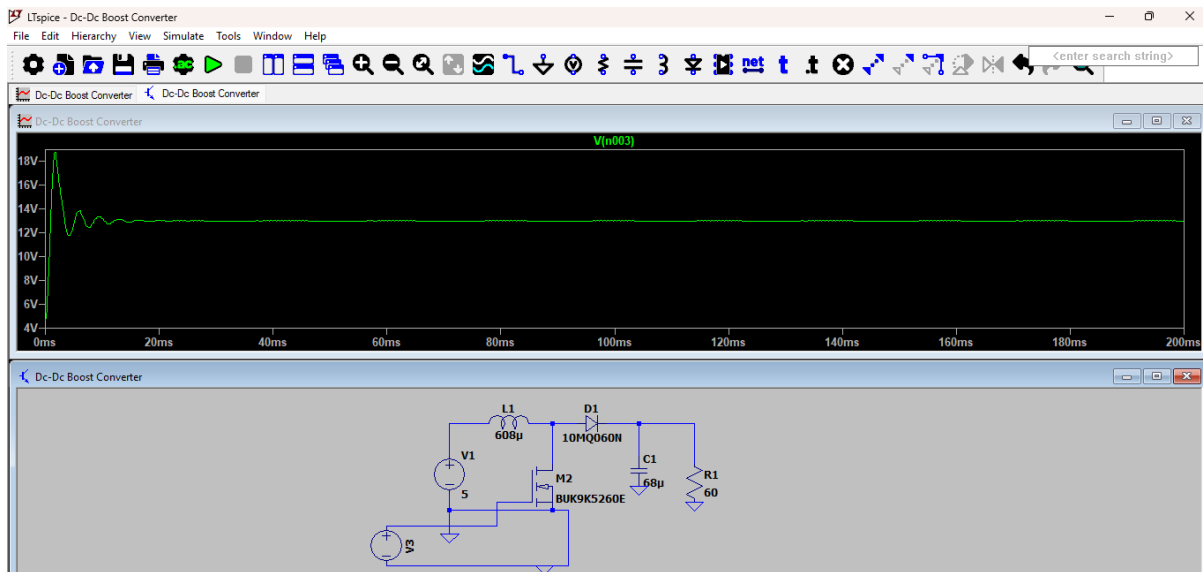
Note: Simulation video for your reference [\(link\)](#)

Expected Output:

Input Voltage: 5V



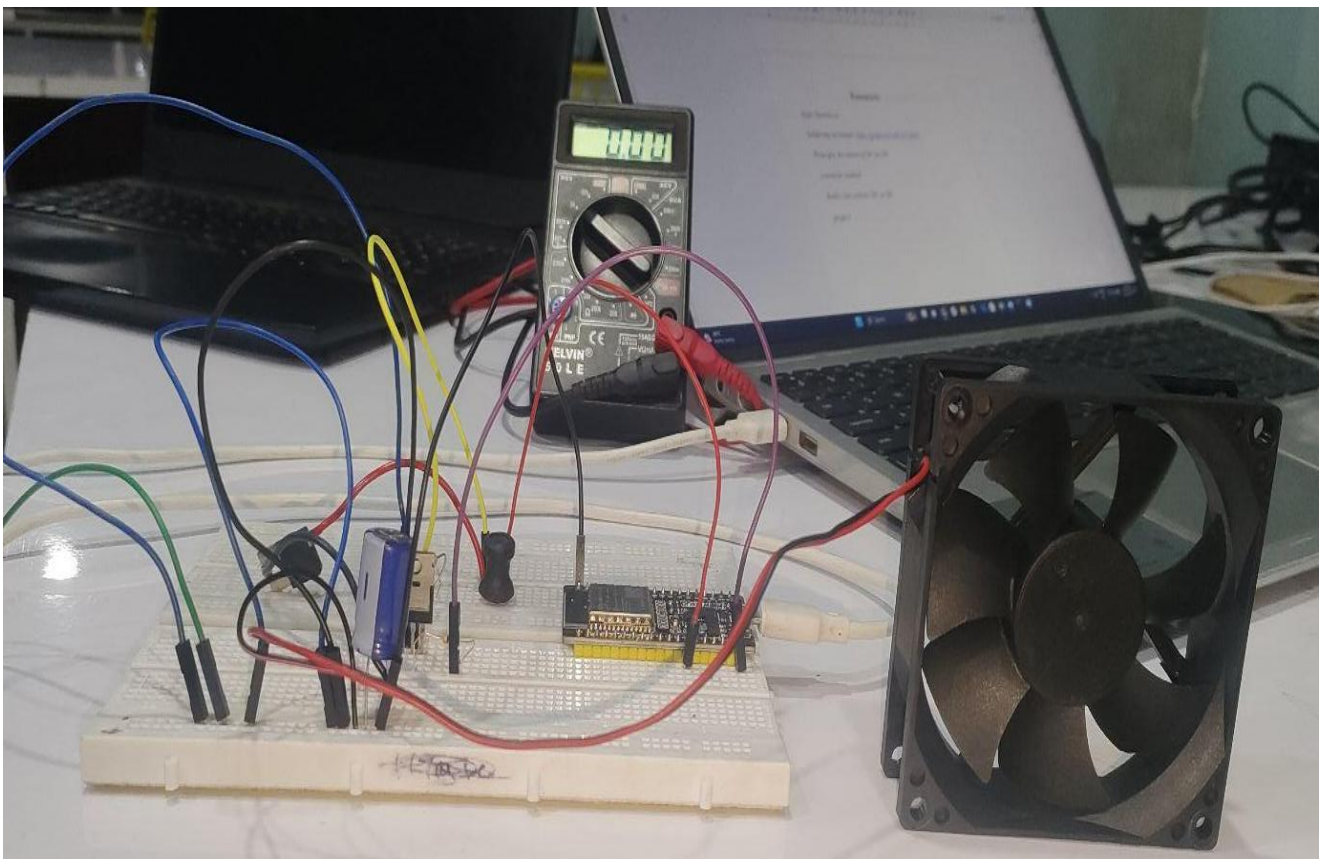
Output Voltage: 12V



Tasks

- 1) Set the input voltage as 12V and design a Boost Converter to get an output of around 24V.
- 2) Simulate the same Boost Converter circuit with different load resistances (e.g., 60Ω, 120Ω, 180Ω) and observe how the output voltage changes.
- 3) Change the inductor value (e.g., 608μH to 300μH or 900μH) and observe how it affects the ripple in output voltage and current.
- 4) Increase or decrease the switching frequency and note how the output voltage ripple and efficiency are affected.
- 5) Plot and observe the Inductor Current, Capacitor Voltage, and MOSFET Drain-Source Voltage during simulation.

Build a simple low power DC-DC Boost Converter using Physical Circuit



Part 2 – Physical design of the circuit

Aim

Components

Connections - Circuit Diagram - Detailed Steps

Software - Launching the IDE for our project - Code

Task - Exercise

Aim

Very often a low power boost converter used to convert some low voltage to a higher voltage is required in a system. This build exercise is to learn to build one such simple converter, using an open loop.

Components

Components	Specification	Cost Per Quantity	Quantity
PCB Dot board	10*15cm	40	1
Inductor	2.2 mH	50	1
Capacitor	100-470uf (63V)	25	1
MOSFET	IRLZ24NPBF OR NVMFS6H818NL	160	1
Schottky Diode	STPSC406D OR NXPSC046506Q OR WNSC6D04650Q	150	1
DC Fan (Load)	12V- 40mA	100	1

Resistor	22,2.2K ohm (Through hole)	15	3 Each one
Gate driver IC	TC4427	100	1
Soldering station (Lead, flux, IP)	Optional	250	1
Power supply	5V 2A Phone Adapter	100	1
Connecting wires	26AWG	20	1(metre)
Bread board	400 pins	50	1
Digital multimeter		175	1
Jumper wire Male to male		15	6
ESP32 Dev Module	C- 6	900	1
USB cable	Type C	200	1

Note:

- All the components are reusable after desoldering.
- Measure the resistor values by multimeter.



Capacitor



Inductor



MOSFET



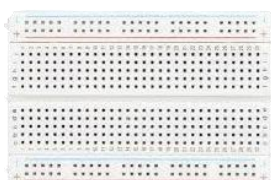
Jumper Wires



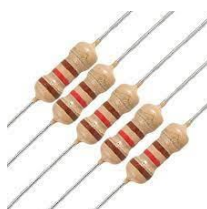
ESP32 C-6 Module



DC Fan



Breadboard



Resistor



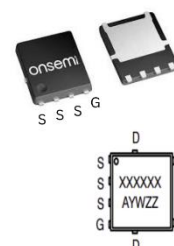
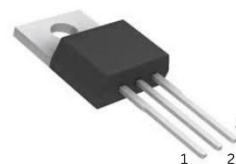
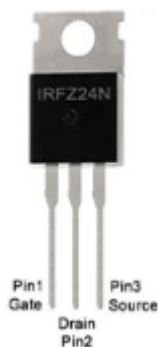
Diode



Gate driver IC

MOSFET PINOUT:

- Pin 1 = gate
- Pin 2 = drain
- Pin 3 = source



Note: While choosing the MOSFET, check from the datasheet that MOSFET V_{gs} should be less than 2.5V.

Connections

Circuit Diagram:

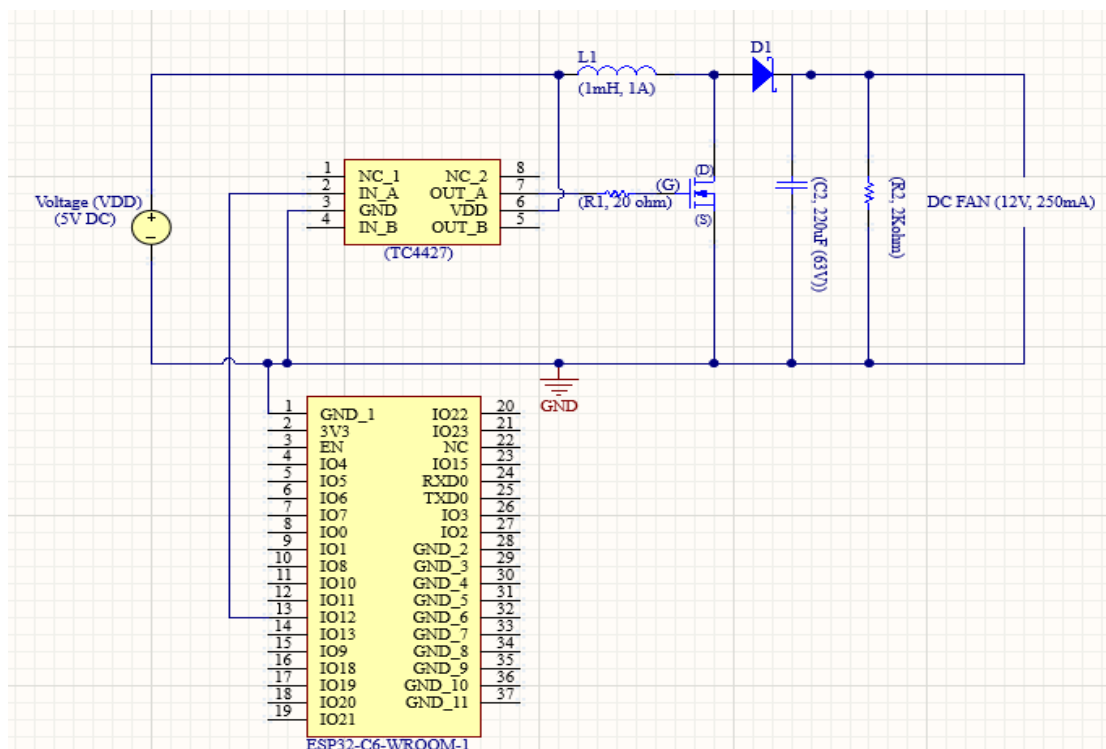
NOTE: First make the circuit diagram in the Bread board and verify the connection properly once you got the output then only go for soldering. Do not touch the tip of the soldering iron, its temperature can be as high as 380°C and can cause severe burns. Keep the cleaning sponge wet when soldering.

Detailed Connection Steps

Step 1: Connect or solder inductor, diode, capacitor, MOSFET, Gate driver IC and resistor on PCB dot board as per the circuit diagram fig1.

Step 2: Set the power supply voltage to 5V or take a 5V phone adapter.

Step 3: Connect or solder positive side (A) of power supply to Inductor leg and negative side of the power supply to the ground (G).



Step 4: Connect or solder another inductor leg(B) to the drain of the MOSFET(D).

- Connect or Solder one leg of the 20-ohm resistor to MOSFET gate and connect another leg of the 20-ohm resistor to pin number 7 (OUT_A) for Gate driver IC.
- Connect or Solder one male to male jumper wire to the MOSFET source and connect another side of jumper wire to GND.

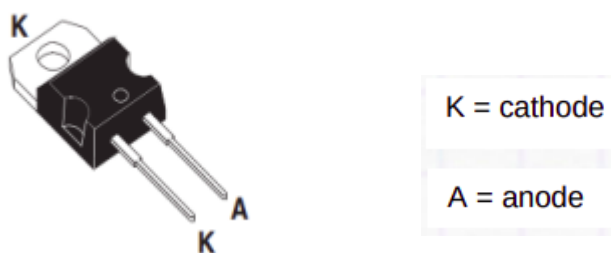
Step 5: Connect or solder pin number 6 (VCC) of the Gate driver IC to Inductor for the input voltage.

Step 6: Connect or solder pin number 2 (IN_A) of the Gate driver IC to pin number 12 of the ESP32.

Step 7: Connect or solder pin number 3 (GND) of the Gate driver IC to GND.

Step 8: Connect or solder drain of the MOSFET to cathode of the diode.

- Connect or solder one leg of the 2K resistor to anode of the diode and connect another leg to GND.
- Connect or solder one leg of the 220uF Capacitor to anode of the diode and connect another leg to GND.
- Connect or solder one common ground to close the circuit.



Step 9: Connect or solder two male to male jumper wire across the capacitor and connect to the 12V DC load fan. K = cathode A = anode DC to DC Boost Converter.

Step 10: Switch ON the power supply and measure the output voltage using digital multimeter across the capacitor.

Step 11: To change the duty cycle, copy the below code and run it in Arduino IDE

Software

Launching the IDE for our project - Code

1. Connect Your ESP32 Module

1. Connect your ESP32 C-6 Module to your computer using a USB cable.
2. Go to Tools > Board > ESP32C6 Dev Module.
3. Go to Tools > Port and select the COM port to which your ESP32C6 is connected.

2. Upload the Code

Copy the provided code into a new sketch in the Arduino IDE.

Click the upload button (right arrow) to compile and upload the code to your ESP32.

Code

“Open the .ino file which was attached in the folder and write a code by the below instruction.”

Main Code

```
#include <Arduino.h>

#include <WiFi.h>

#include <WebServer.h>

// ===== WiFi Credentials =====

const char* ssid = "esp32-c6";

const char* password = "123456789";

// ===== PWM Config =====
```

```

const int pwmPin = 12;

int freq = 113000;

const int ledChannel = 0;

const int resolution = 8;

const int period = 255;

float Dutycycle = 50.0;

int dutyValue;

// ===== Web Server =====

WebServer server(80);

// ===== Functions =====

int calculateDuty(float dutyPercent) {

    return (period * (dutyPercent / 100.0f));

}

void Set_Duty(int dutyValue) {

    if (dutyValue >= 0 && dutyValue <= period) {

        ledcWrite(ledChannel, dutyValue);

    }

}

// ===== HTML Page with Slider & Mobile View =====

String htmlPage() {

    String page = "<!DOCTYPE html><html><head>";
  
```

```

page += "<meta name='viewport' content='width=device-width, initial-
scale=1.0'>";

page += "<title>ESP32 PWM Control</title>";

page += "<style>";

page += "body{font-family:Arial;text-align:center;padding:10px;}";

page += "input[type=number], .slider{width:80%; max-width:300px; margin:10px
0;}";

page += "h2, h3{margin:10px 0;}";

page += "button{padding:10px 20px;font-size:16px;margin:10px 0;}";

page += "</style></head><body>";

page += "<h2>ESP32 PWM Control</h2>";

// Frequency input + submit button

page += "<h3>Frequency Control</h3>";

page += "<input type='number' id='freqInput' value='" + String(freq) + "'>";

page += "<br><button onclick='updateFreq()'>Update
Frequency</button><br><hr>";

// Duty cycle slider

page += "<h3>Duty Cycle Control</h3>";

page += "<input type='range' min='0' max='100' value='" + String((int)Dutycycle) +
'" class='slider' id='pwmSlider' oninput='updateDuty(this.value)'>";

```

```

page += "<p>Duty Cycle: <span id='dutyVal'>" + String((int)Dutycycle) +
"</span>%</p>";

// JS AJAX for slider and frequency

page += "<script>";

// Duty cycle slider

page += "function updateDuty(val){";

page += "document.getElementById('dutyVal').innerText=val;";

page += "var xhr=new XMLHttpRequest();";

page += "xhr.open('GET','/set?duty='+val,true);";

page += "xhr.send();}";

// Frequency update

page += "function updateFreq(){";

page += "var f=document.getElementById('freqInput').value;";

page += "var xhr=new XMLHttpRequest();";

page += "xhr.open('GET','/set?freq='+f,true);";

page += "xhr.send();}";

page += "</script>";

page += "</body></html>";

return page;

}

// ===== Web Handlers =====

```

```
void handleRoot() {  
    server.send(200, "text/html", htmlPage());  
}  
  
void handleSet() {  
    if (server.hasArg("freq")) {  
        freq = server.arg("freq").toInt();  
        ledcChangeFrequency(ledChannel, freq, resolution);  
    }  
  
    if (server.hasArg("duty")) {  
        Dutycycle = server.arg("duty").toFloat();  
        dutyValue = calculateDuty(Dutycycle);  
        Set_Duty(dutyValue);  
    }  
  
    server.send(200, "text/plain", "OK");  
}  
  
// ===== Setup =====  
  
void setup() {  
    Serial.begin(115200);  
  
    // WiFi AP mode  
    WiFi.softAP(ssid, password);  
  
    Serial.print("AP IP Address: ");
```



```
Serial.println(WiFi.softAPIP());

// Web server routes

server.on("/", handleRoot);

server.on("/set", handleSet);

server.begin();

// PWM setup (preserve your flow)

ledcAttach(ledChannel, freq, resolution);

ledcAttachChannel(pwmPin, freq, resolution, 0);

dutyValue = calculateDuty(Dutycycle);

Set_Duty(dutyValue);

Serial.print("PWM started on GPIO ");

Serial.print(pwmPin);

Serial.print(" with duty cycle = ");

Serial.print(dutyValue);

Serial.println(" / 255");

}

// ===== Loop =====

void loop() {

    server.handleClient();

}
```

Hurray you have learnt how to drive a boost circuit!!!!!!

Tasks:

For open loop

1. On the same boost converter change the duty cycle 50%, 60% and 70% and measure the voltage.
2. Measure the output voltage keeping $R = 12\Omega$. Compute the expected voltage using mathematical formula for each of the duty cycle. Check measure voltage and compute voltage are matching.

Exercise:

1. Calculate the losses and the efficiency using the formula.
2. Make a graph
 - between ripple voltage and duty
 - between ripple voltage and load